Project: To-Do List API **CR ID:** CR-005 — AI-Assisted Task Breakdown **All statuses below are initial. No test has been executed.**

Test ID	Requirement ID	Test Scenario	Expected Result	Status
TC-005-01	FR-01, FR-02	Submit a task with a description containing several distinct activities	`200 OK`; `shouldBreakDown: true`; `suggestedTasks` contains at least 2 items, each with `title`, `description`, `dueDate`	Not Executed
TC-005-02	FR-02, FR-03	Submit a small, atomic task (e.g. "Call the dentist")	`200 OK`; `shouldBreakDown: false`; `reason` explains why; `suggestedTasks` is empty	Not Executed
TC-005-03	FR-04	After TC-005-01, inspect the `TASK` table	No new rows were created as a result of the breakdown call	Not Executed
TC-005-04	FR-01	Submit a request with `title` missing	`400 Bad Request`	Not Executed
TC-005-05	FR-06	Simulate the AI provider being unreachable (timeout)	`502 Bad Gateway` with a clear message; no task row created	Not Executed
TC-005-06	FR-06	Simulate the AI provider returning malformed/unparsable JSON	`502 Bad Gateway`; treated as a provider failure, not as "no breakdown needed"	Not Executed
TC-005-07	FR-06	While the AI provider is simulated as down (TC-005-05 conditions), call `POST /api/tasks` normally	`201 Created`; ordinary task creation is completely unaffected	Not Executed
TC-005-08	FR-05	Inspect the database schema after this CR is implemented	No `ParentTask` / `Subtask` entity or parent-child relationship exists	Not Executed
TC-005-09	FR-07	Take a suggested task from TC-005-01's response, submit it via `POST /api/tasks` with its suggested `dueDate` changed to a past date	`400 Bad Request`, same validation as any other task creation (`@FutureOrPresent`)	Not Executed
TC-005-10	FR-07	Take a suggested task from TC-005-01's response and submit it via `POST /api/tasks` unmodified	`201 Created`; a real, independent task now exists, with no link back to the original task or the other suggestions	Not Executed
TC-005-11	§6 step 5 (safeguard)	Simulate the AI returning a suggested due date after the original task's due date	That suggestion is filtered out of the response rather than causing the whole call to fail	Not Executed
TC-005-12	FR-08	Inspect application logs and API responses after a provider call	No AI provider API key or credential appears in logs or in any response body	Not Executed
TC-005-13	— (regression)	Run the existing Playwright `tasks.spec.ts` and `categories.spec.ts` suites unmodified	All existing test cases continue to pass; existing endpoints are untouched by this CR	Not Executed

Notes

- TC-005-05, TC-005-06, and TC-005-07 should use a mocked/stubbed AI provider in the test environment rather than depending on a real external service's availability, so these tests are deterministic and do not depend on network access to a third-party provider.